# PRACTICAL 8

## MySQL Schema Design, ER Model, PDO Connectivity and Prepared Statements

### AIM

To design a normalized StudentHub database with students, events and registrations, define primary and foreign keys, connect PHP securely using PDO, and retrieve records using a prepared statement.

### Introduction

StudentHub was extended with a relational database named studenthub_db. Students and events are stored separately, while registrations links each student to an event. The database was populated with sample records and accessed through PHP using a dedicated database account. Browser tests confirmed the connection and demonstrated valid, empty and invalid-input results.

### Tools and Technology

VS Code, XAMPP, MariaDB 10.4.32, phpMyAdmin 5.2.1, PHP 8.2.12, PDO MySQL driver, SQL, browser, Git and GitHub. MariaDB is the MySQL-compatible database server supplied with this XAMPP installation.

### Files Used

| File | Purpose |
|---|---|
| sql/studenthub_db.sql | Database structure, seed data, indexes and constraints. |
| php/db.php | Reusable PDO connection returned to calling scripts. |
| php/db-config.php | Local database settings and password; excluded from Git. |
| php/db-config.example.php | Configuration template containing a password placeholder. |
| php/test-connection.php | Connection test, database user and table counts. |
| php/prepared-demo.php | Validated student ID and parameterized registration query. |
| .gitignore | Excludes the private configuration and Practical 7 storage. |

### Key Skills Used

Relational schema design, normalization, primary and foreign keys, UNIQUE and CHECK constraints, SQL JOINs, seed data, database privileges, PDO exception handling and prepared statements.

## Implementation Overview

### 1 Database and Tables

The database uses InnoDB tables and utf8mb4_unicode_ci collation. Each table has an unsigned integer primary key with AUTO_INCREMENT. Students contain contact and study details; events contain the title, description, date and venue.

### 2 Relationships and Constraints

registrations.student_id references students.student_id, and registrations.event_id references events.event_id. Updates cascade to the referencing keys; deleting a referenced parent is restricted. A unique student/event pair prevents duplicate event signups, unique student email prevents duplicates, and chk_student_year limits the year to 1 through 4.

### 3 Normalization and ER Model

Each column stores a single value, and each table represents one entity or relationship. Student and event details are kept in their own tables rather than repeated for every registration. The registrations table resolves the many-to-many relationship into two one-to-many relationships. In this practical, registrations means event signups, separate from the account requests stored in Practical 7.

### 4 Seed Data and JOIN Query

Five students, three events and six registrations were inserted. An INNER JOIN across the three tables displayed each registration with its student name, event title, date and venue.

### 5 PDO Connectivity

A dedicated studenthub_app account was granted SELECT, INSERT, UPDATE and DELETE on studenthub_db. db.php reads local configuration and creates a PDO connection using charset=utf8mb4, exception error mode, associative fetch mode and disabled emulated prepares. The test page confirmed the database account and all three record counts.

### 6 Prepared Statement and Error Handling

prepared-demo.php validates student_id as a positive integer, prepares a JOIN query containing :student_id, binds the value using PDO::PARAM_INT, and executes it. Invalid input is rejected before the query. Exceptions are caught; a general browser message is returned and technical details are logged on the server.

### 7 SQL Export and Source Submission

The SQL dump includes structure, seed records, indexes and foreign keys. The students definition was corrected in the export to retain its CHECK constraint, engine and character settings. The corrected text was reviewed; a fresh import was not tested. Source files were pushed to GitHub in commit 0ee6a87.

## Database Structure and ER Model

| Table | Primary key | Other constraints | Rows |
|---|---|---|---|
| students | student_id | Unique email; year CHECK 1–4 | 5 |
| events | event_id | Required title, date and venue | 3 |
| registrations | registration_id | Two foreign keys; unique student/event pair | 6 |

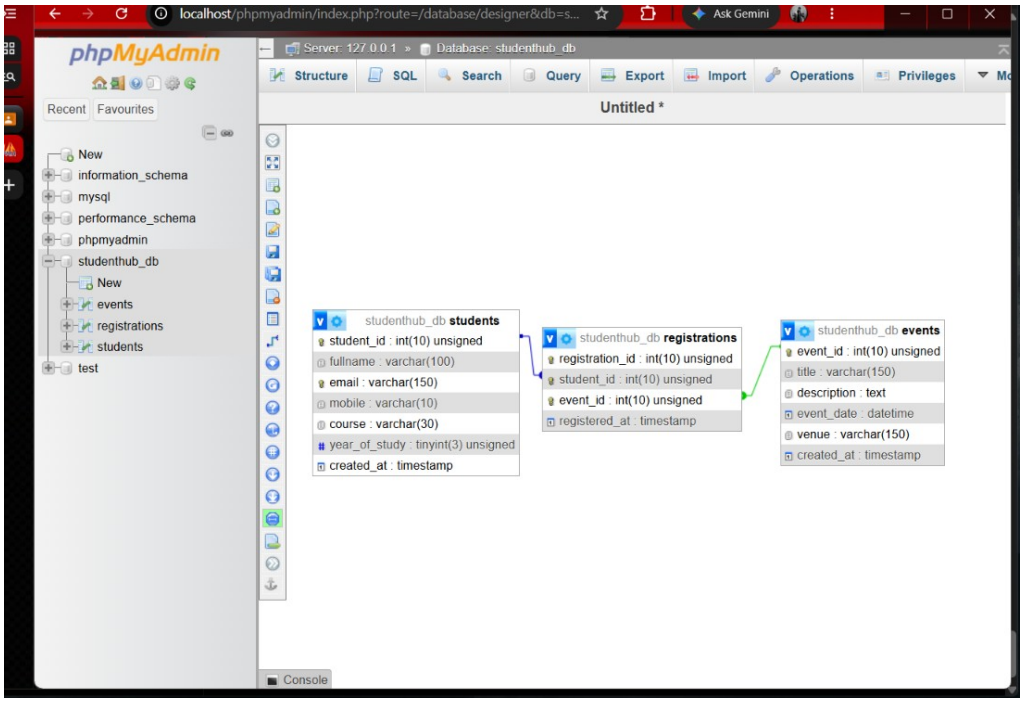



Figure 1: phpMyAdmin Designer showing students and events connected through registrations.

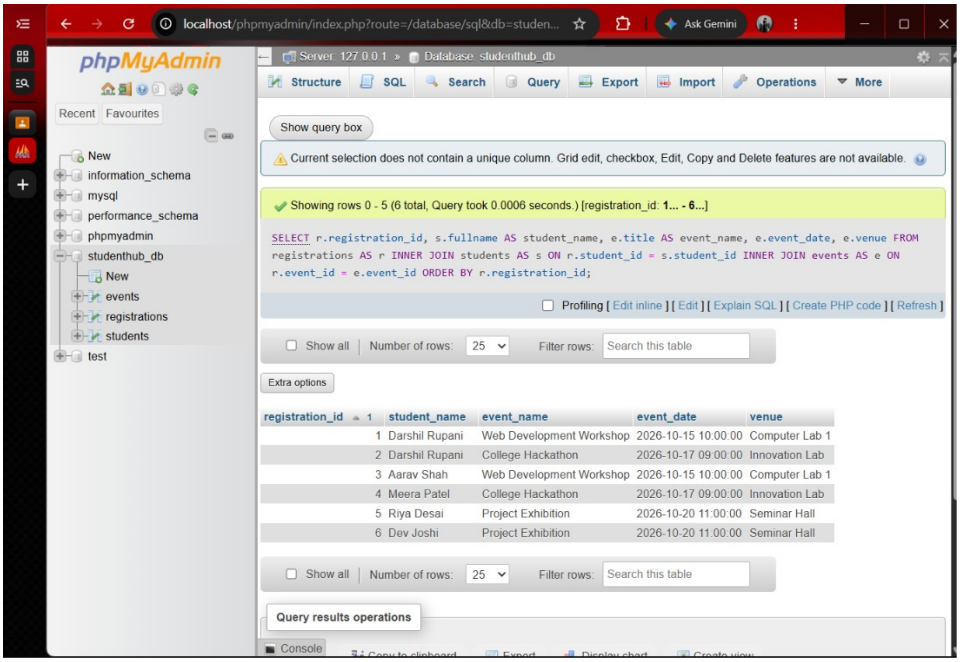



Figure 2: JOIN query displaying all six registrations with student and event details.

## Screenshots of PDO and Prepared Statement Results

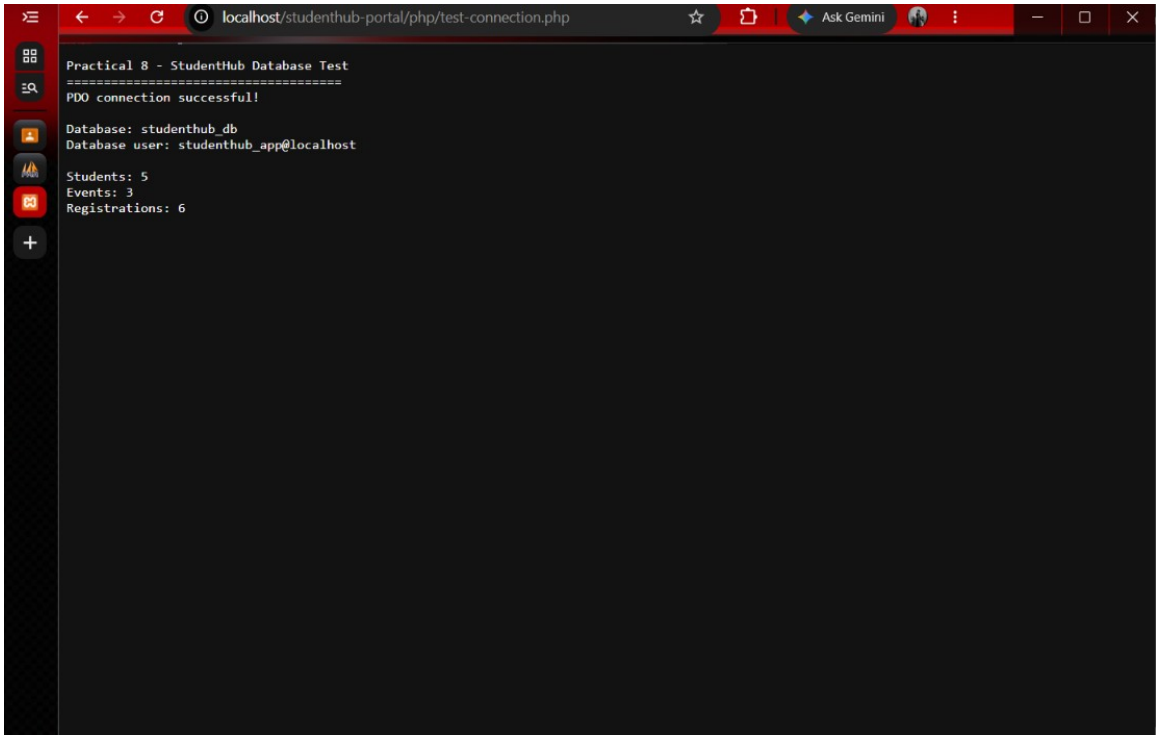



Figure 3: Successful PDO connection using studenthub_app, with 5 students, 3 events and 6 registrations.

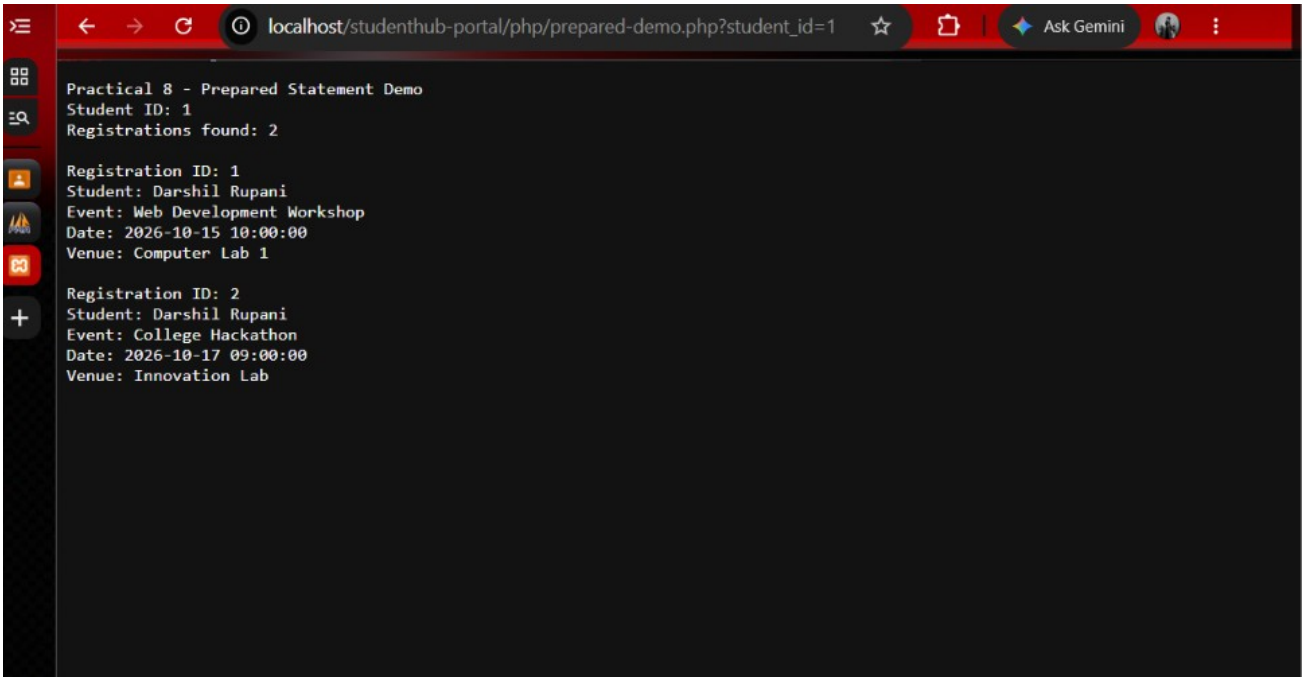



Figure 4: student_id=1 returns two event registrations for Darshil Rupani.

Connection test URL: http://localhost/studenthub-portal/php/test-connection.php

Prepared-statement URL: http://localhost/studenthub-portal/php/prepared-demo.php?student_id=1

## Screenshots of Additional Test Cases

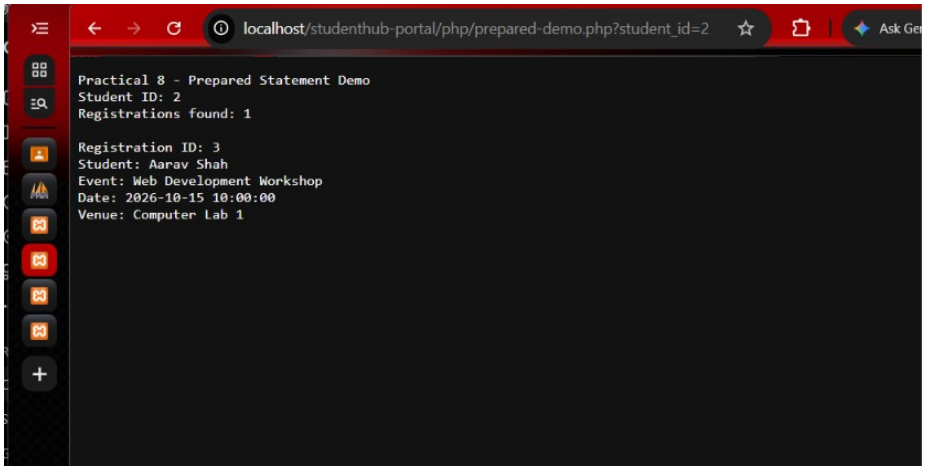



Figure 5: student_id=2 returns one registration for Aarav Shah.

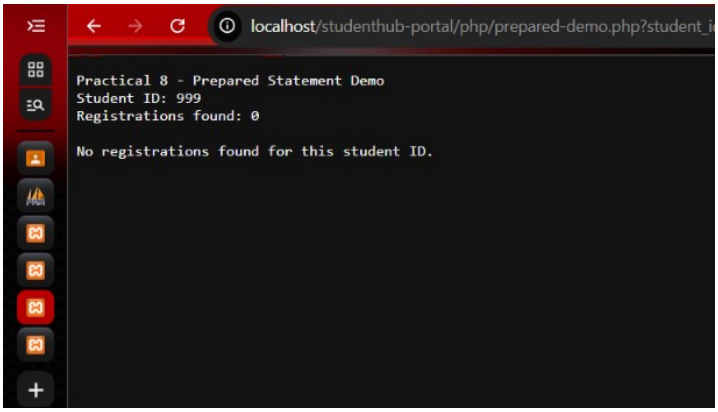



Figure 6: student_id=999 returns zero registrations and a clear message.

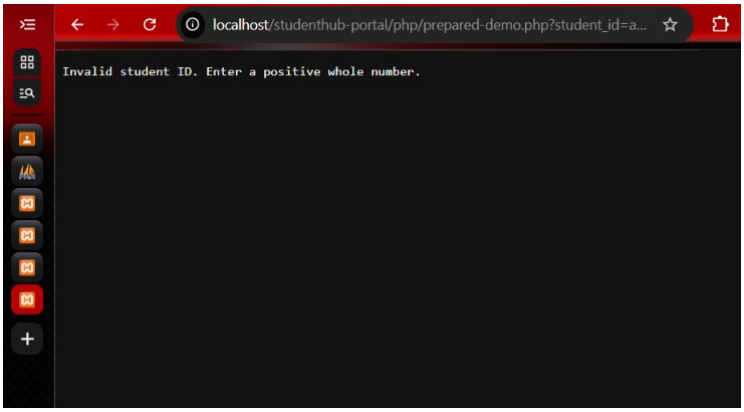



Figure 7: Non-numeric input is rejected with an invalid student ID message.

## Testing Summary

| Test input or action | Observed result |
|---|---|
| PDO connection test | Successful; counts: students 5, events 3, registrations 6. |
| student_id=1 / student_id=2 | Two registrations / one registration returned. |
| student_id=999 | Zero registrations with a no-records message. |
| Non-numeric student_id | Rejected: enter a positive whole number. |

## Key Questions and Answers

### Q1 Is the schema normalized properly

The design separates student details, event details and the association between them. Values are atomic, and descriptive attributes depend on the key of their own entity. The registration table stores references rather than repeating names and event descriptions. This structure supports third normal form for the modeled data.

### Q2 Are primary keys and foreign keys correctly defined

Each table has an AUTO_INCREMENT primary key. Both registration foreign keys reference the matching unsigned integer parent keys. The ER diagram shows both relationships. A UNIQUE constraint on student_id and event_id prevents duplicate signups for the same event.

### Q3 Is the database connection handled securely

PHP connects using the dedicated studenthub_app account rather than root. The password is kept in db-config.php, which Git ignores; the shared example contains a placeholder. PDO exceptions are caught by the calling pages, and detailed errors are logged rather than shown in the browser.

### Q4 Why use a prepared statement

The query structure is prepared separately from the supplied value. The student ID is bound to :student_id as an integer, so it is treated as a parameter rather than concatenated SQL. Positive-integer validation also provides a clear response for unsuitable input.

## GitHub Repository

https://github.com/DarshilRupani/studenthub-portal

Practical 8 source commit: 0ee6a87
Message: Complete Practical 8 database schema and PDO prepared statements
Push to main was confirmed from terminal output. A later report update was pushed as 82dac4a; the working tree was clean at that point.

## Post Laboratory Work

Submit sql/studenthub_db.sql, php/db.php, the supporting PHP scripts and the connection-test screenshot with this report. To reproduce the setup, import the dump into an empty studenthub_db database, create the application user with the required privileges, and copy db-config.example.php to db-config.php with local credentials.

## Conclusion

The StudentHub database was designed and populated successfully. Its ER model represents the student-to-event relationship through registrations. PDO connectivity and parameterized retrieval were demonstrated with successful, empty and invalid-input cases. The intermediate PDO and prepared-statement extension was completed. The optional stored-procedure extension was not implemented.

Verification scope: screenshots confirm the browser results. A fresh dump import, deliberate constraint violations and a forced connection failure were not tested.